

Jiseok Chae | 채지석

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Education

KAIST

Integrated M.S./Ph.D. Program, Mathematical Sciences Mar. 2021 – Feb. 2026

GPA : 4.24/4.3

Thesis : Two Theoretical Advances in Generative Models: minimax optimization methods with improved convergence and provable benign overfitting

Advisor : Prof. Donghwan Kim

KAIST

Bachelor of Science, Mathematical Sciences

Mar. 2016 – Feb. 2021

Graduated *summa cum laude*, GPA 4.06/4.3

Minor in Computer Science

Professional Interests

- Continuous Optimization for Machine Learning
- Design and Analysis of Machine Learning Algorithms
- Generative Models and Understanding How They Work

Publications

Selected Works.....

Stochastic Extragradient with Flip-Flop Shuffling & Anchoring: Provable Improvements

Jiseok Chae, Chulhee Yun, Donghwan Kim

38th Annual Conference on Neural Information Processing Systems

NeurIPS 2024

Two-timescale Extragradient for Finding Local Minimax Points

Jiseok Chae, Kyuwon Kim, Donghwan Kim

Twelfth International Conference on Learning Representations

ICLR 2024

HEaN-STAT: a privacy-preserving statistical analysis toolkit for large-scale numerical, ordinal, and categorical data

Younho Lee, Jinyeong Seo, Yujin Nam, Jiseok Chae, Jung Hee Cheon

IEEE Transactions on Dependable and Secure Computing, Issue 3, 2024

TDSC 2024

Peer-reviewed.....

Optoelectronic Synaptic Transistors Based on Colloidal CdSe Nanowires for Energy-Efficient Neuromorphic Computing

Woosik Kim, Jiseok Chae*, Taesung Park, Hanseok Seo, Hyung Jin Choi, Byung Ku Jung, Seungshin Lim, and Soong Ju Oh*

*equal contribution

ACS Applied Materials & Interfaces, 18(5), 2026

Open Problem: Is There a First-Order Method that Only Converges to Local Minimax

Optima?

Jiseok Chae, Kyuwon Kim, Donghwan Kim

36th Annual Conference on Learning Theory (Open Problems Track)

COLT 2023

Workshops/Preprints.....

Rethinking Memorization–Generalization Trade–Off in Generative Models

Jiseok Chae*, Kyuwon Kim*, Donghwan Kim

*equal contribution

3rd Workshop on High-dimensional Learning Dynamics, ICML 2025

Professional Experience

CryptoLab, Inc.

Seoul

Research Programmer Internship

Jun. 2019 – Aug. 2019

- Developed HEaaN.STAT, a statistical data analysis program using the homomorphic encryption scheme HEaaN. Joint work with Korea Credit Bureau

Skills

(Programming) Languages & Software

- Python (NumPy, PyTorch, ...)
- C++
- MATLAB
- L^AT_EX

(Natural) Languages

- Korean (Native)
- English (Fluent)
- Japanese (Intermediate)

Invited/Contributed Talks

Rethinking Memorization–Generalization Trade–Off in Generative Models

2025 Korean Mathematical Society Annual Meeting; Korea, 2025

Stochastic Extragradient With Flip-Flop Shuffling & Anchoring

2025 International Conference on Continuous Optimization; USA, 2025

2024 Korean Mathematical Society Spring Meeting; Korea, 2024

Two-timescale Extragradient for Finding Local Minimax Points

KAIST Stochastic Analysis and Application Research Center ColLabor Workshop; Korea, 2023

Selected Honors and Awards

Outstanding Paper Award

KAIST

Doctoral dissertation

2026

National Academic Excellence Scholarship (Science and Engineering)

Korea Student Aid Foundation (KOSAF)

2016 – 2020

Dean’s List

College of Natural Sciences, KAIST

Spring 2019

School of Freshman, KAIST

Spring 2016

Teaching Experience

KAIST

- Head Teaching Assistant
 - MAS110 Linear Algebra for Data Science *Spring 2022 – Spring 2024*
 - The founding head TA of the newly established course.
- Programming Teaching Assistant
 - MAS109 Introduction to Linear Algebra *Spring 2022 – Spring 2024*
 - Taught MATLAB; designed course materials and programming assignments.
- Teaching Assistant
 - MAS.10002 Calculus II *Fall 2025*
 - MAS473 Introduction to AI with Mathematics *Fall 2024*
 - MAS374 Optimization Theory *Fall 2021*
 - MAS101 Calculus I *Spring 2021 – Fall 2021*

Additional Information

- Technical Research Personnel (전문연구요원) *Mar. 2024 – Present*
- *Regular member* of the Korea Cube Culture United (KCCU)